Supplementary Information

Mechanistic Monte Carlo assessment of co-localized permeation enhancement and efflux modulation for oral peptide delivery

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Code and data: https://github.com/sandlerleon/dual-lock-oral-peptide-mc (software DOI 10.5281/zenodo.23864330; preprint DOI 10.5281/zenodo.23864327)

This file contains the parameter provenance, the complete arm results, the full sensitivity indices, the structural variants, the convergence studies, the list of verification tests and the literature values used in the plausibility checks. All numbers are generated from the result files in the repository (results/ext_*.json, results/tables/*.csv).

Table S1. Parameter priors and provenance

Symbol	Unit	Grade	Prior	Basis
P0	cm/h	C	lognormal median 0.009746 (calibrated), sigma_ln 0.6	Calibrated so that the median of the reference arm R1 equals 1.2 % (mid-point of the 1-2 % enhancer-tablet range, Brayden et al. 2020; oral semaglutide 0.8 %, Overgaard et al. 2021). Dispersion: author assumption.
k_deg	1/h	D	lognormal median 2.0, sigma_ln 0.5	Author assumption (no direct measurement for an enhancer niche); order of 1 h^-1 for luminal proteolysis of an unprotected peptide.
S_post	-	D	beta mean 0.55, sd 0.15	Author assumption (intra-enterocyte handling and first pass).
ER0	-	C	lognormal median 2.5, sigma_ln 0.5	Caco-2 bidirectional efflux ratios for P-glycoprotein substrates, order of magnitude (Collnot et al. 2006); applies only if the peptide is a substrate.
p_sub	-	D	scenario axis, base 0.5	Scenario axis, not a prior (peptide-specific; unknown for any given 2-4 kDa peptide).
A_m	cm2	D	lognormal median 30.0, sigma_ln 0.3	Author assumption: mucosal contact area of the tablet niche.
EC50	mM	C	lognormal median 25.0, sigma_ln 0.5	Millimolar range of SNAC and sodium caprate effective concentrations (Maher et al. 2016; Twarog et al. 2019).
Emax	-	D	lognormal median 80.0, sigma_ln 0.5	Author assumption (maximal fold-increase of peptide permeability).
h	-	D	lognormal median 2.0, sigma_ln 0.25	Author assumption (Hill coefficient).
C_tox	mM	D	lognormal median 80.0, sigma_ln 0.4	Author assumption (half-damage concentration of the enhancer).
t_rev	h	D	lognormal median 1.0, sigma_ln 0.5	Author assumption (epithelial recovery time after enhancer clearance).
IC50	mM	C	lognormal median 0.5, sigma_ln 0.8	Piperine 15-74 uM in Caco-2 (Bhardwaj et al. 2002); TPGS sub-millimolar (Collnot et al. 2006); wide dispersion.

Table S1. *Common priors with evidence grade and basis (grade A, direct measurement; B, close analogue; C, order-of-magnitude literature range; D, author assumption). No input reaches grade A. Eight of twelve common inputs are grade D and four are grade C.*

Arm	PE (mmol)	Modulator (mmol)	δ (cm)	k_{dil} (h^{-1})	m	Pr	k_{dep} (h^{-1})	k_{rel} (h^{-1})
R0_peptide_alone	0	0	1	1.5	0.6	0	0.5	2
R1_reference_PE	1	0	1	1.5	0.5	0.5	0.5	2
R2_reference_PE_efflux	1	0.033	1	1.5	0.5	0.5	0.5	2
P1_dual_lock	0.5	0.033	0.3	0.8	0.15	0.7	3	1.2
P2_no_efflux_module	0.5	0	0.3	0.8	0.15	0.7	3	1.2
P3_no_colocalisation	0.5	0.033	1	1.5	0.5	0.7	3	2
P4_no_protection	0.5	0.033	0.3	0.8	0.15	0.5	3	1.2
P5_no_self_limiting_PE	0.5	0.033	0.3	0.8	0.15	0.7	0.5	1.2

Table S1b. *Arm definitions (medians of the formulation-controlled inputs; all grade D).*

Table S2. Complete arm results (paired draws)

Arm	Median F	F 5%	F 95%	P(F≥4%)	P(reversible)	P(success)	Δ vs R1	Δ vs P1
R0_peptide_alone	0.09	0.02	0.4	0.0	99.7	0.0	-15.0	-45.0
R1_reference_PE	1.19	0.09	11.0	20.9	83.7	15.0	+0.0	-30.0

Arm	Median *F*	*F* 5%	*F* 95%	P(*F*≥4%)	P(reversible)	P(success)	Δ vs R1	Δ vs P1
R2_reference_PE_efflux	1.38	0.11	12.0	23.4	83.7	16.8	+1.9	-28.1
P1_dual_lock	4.95	0.57	28.5	56.3	85.9	45.0	+30.0	+0.0
P2_no_efflux_module	3.88	0.39	25.9	49.2	85.9	38.8	+23.8	-6.2
P3_no_colocalisation	0.35	0.05	3.1	3.4	98.8	3.3	-11.7	-41.7
P4_no_protection	4.20	0.46	25.9	51.5	85.9	40.7	+25.7	-4.3
P5_no_self_limiting_PE	13.23	1.40	43.0	82.4	23.8	14.1	-0.8	-30.8
M0_matrix_only	0.77	0.14	3.3	3.1	99.7	3.0	-11.9	-41.9
R1_half_dose	0.47	0.05	5.4	7.7	93.5	6.4	-8.5	-38.5
P1_full_dose	10.69	1.13	38.7	78.0	67.1	48.2	+33.2	+3.2
P1_minus_delta	0.45	0.07	3.3	3.8	98.0	3.5	-11.5	-41.5
P1_minus_k_dil	3.08	0.34	21.5	42.5	90.4	35.7	+20.7	-9.3
P1_minus_mism	4.76	0.55	27.7	55.1	84.3	43.0	+28.1	-1.9
P1_minus_k_rel	7.36	0.79	34.4	67.5	94.5	62.9	+47.9	+18.0
R1_plus_confinement_bundle	15.28	2.00	44.5	87.9	7.6	4.7	-10.2	-40.2
R1_plus_delta	12.08	1.43	39.8	82.6	49.5	38.0	+23.1	-6.9
R1_plus_k_dil	2.17	0.16	16.8	34.1	59.3	14.1	-0.9	-30.8
R1_plus_mism	1.24	0.09	11.2	21.5	86.3	16.0	+1.0	-29.0
R1_plus_k_rel	0.79	0.07	8.3	14.5	72.2	7.2	-7.8	-37.8
R1_plus_prot	1.37	0.10	12.4	23.7	83.7	17.1	+2.1	-27.8
R1_plus_k_dep	0.53	0.05	5.6	8.4	97.2	7.8	-7.1	-37.1
R1_plus_halfPE_efflux	0.55	0.06	6.0	9.0	93.5	7.5	-7.5	-37.5

Table S2. *All arms of the paired design (20,000 shared draws). Percentages; Δ is the paired change in P(success) relative to R1 or P1 in percentage points.*

Table S3. Sobol sensitivity indices

Input	S_1	95% CI	S_T	95% CI
delta	0.222	[0.183, 0.267]	0.242	[0.231, 0.251]
EC50	0.172	[0.143, 0.204]	0.191	[0.184, 0.199]
P0	0.176	[0.149, 0.210]	0.182	[0.175, 0.188]
A_m	0.073	[0.050, 0.097]	0.082	[0.079, 0.086]
h	0.060	[0.039, 0.086]	0.076	[0.072, 0.079]
Emax	0.061	[0.037, 0.082]	0.066	[0.063, 0.069]
S_post	0.066	[0.045, 0.089]	0.064	[0.062, 0.066]
k_dep	0.032	[0.017, 0.048]	0.037	[0.035, 0.038]
sub	0.022	[0.007, 0.035]	0.036	[0.033, 0.038]
k_dil	0.022	[0.011, 0.034]	0.022	[0.021, 0.023]
ER0	0.009	[-0.003, 0.021]	0.019	[0.018, 0.021]
k_rel	0.012	[0.002, 0.023]	0.016	[0.016, 0.017]
k_deg	0.010	[0.001, 0.019]	0.013	[0.013, 0.014]
IC50	0.002	[-0.005, 0.009]	0.008	[0.007, 0.008]
Pr	0.006	[-0.001, 0.012]	0.006	[0.006, 0.007]
m	0.002	[-0.002, 0.008]	0.003	[0.003, 0.003]
C_tox	0.000	[0.000, 0.000]	0.000	[0.000, 0.000]
t_rev	0.000	[0.000, 0.000]	0.000	[0.000, 0.000]

Table S3a. *First-order (S_1) and total-order (S_T) indices for log $_{10}F$, P1 Dual-Lock, with 95% bootstrap intervals (N=2^{13} base samples).*

Input	S_1	95% CI	S_T	95% CI
delta	0.192	[0.148, 0.245]	0.204	[0.196, 0.211]
EC50	0.174	[0.138, 0.216]	0.188	[0.181, 0.195]
P0	0.152	[0.124, 0.187]	0.152	[0.146, 0.157]
sub	0.092	[0.057, 0.124]	0.117	[0.112, 0.121]
Emax	0.071	[0.041, 0.098]	0.073	[0.070, 0.076]
A_m	0.063	[0.038, 0.088]	0.069	[0.066, 0.072]
h	0.039	[0.016, 0.067]	0.056	[0.054, 0.059]
ER0	0.028	[0.005, 0.047]	0.052	[0.049, 0.054]
k_dil	0.050	[0.030, 0.072]	0.050	[0.048, 0.052]

Input	S_1	95% CI	S_T	95% CI
S_post	0.045	[0.025, 0.068]	0.044	[0.042, 0.045]
m	0.022	[0.005, 0.038]	0.026	[0.025, 0.028]
k_deg	0.012	[0.002, 0.023]	0.013	[0.013, 0.014]
k_rel	0.006	[-0.003, 0.016]	0.009	[0.009, 0.010]
Pr	0.002	[-0.003, 0.006]	0.002	[0.002, 0.002]
k_dep	0.002	[-0.003, 0.006]	0.002	[0.002, 0.002]
C_tox	0.000	[0.000, 0.000]	0.000	[0.000, 0.000]
t_rev	0.000	[0.000, 0.000]	0.000	[0.000, 0.000]
IC50	0.000	[0.000, 0.000]	0.000	[0.000, 0.000]

Table S3b. *First-order (S_1) and total-order (S_T) indices for log $_{10}$F, R1 reference, with 95% bootstrap intervals (N=2^{13} base samples).*

Arm	Output	Group	Closed	Total	95% CI (total)
P1	log10 F	peptide and epithelium	0.369	0.386	[0.373, 0.398]
P1	log10 F	enhancer pharmacology	0.294	0.323	[0.312, 0.335]
P1	log10 F	efflux modulator	0.002	0.008	[0.007, 0.008]
P1	log10 F	formulation architecture	0.300	0.320	[0.310, 0.331]
P1	success	peptide and epithelium	0.162	0.550	[0.532, 0.568]
P1	success	enhancer pharmacology	0.230	0.680	[0.660, 0.696]
P1	success	efflux modulator	-0.002	0.043	[0.038, 0.049]
P1	success	formulation architecture	0.134	0.561	[0.542, 0.582]
P1	log10 DH	peptide and epithelium	0.228	0.228	[0.221, 0.235]
P1	log10 DH	enhancer pharmacology	0.404	0.404	[0.391, 0.415]
P1	log10 DH	efflux modulator	0.000	0.000	[0.000, 0.000]
P1	log10 DH	formulation architecture	0.369	0.369	[0.358, 0.378]
R1	log10 F	peptide and epithelium	0.418	0.423	[0.409, 0.435]
R1	log10 F	enhancer pharmacology	0.287	0.309	[0.299, 0.317]
R1	log10 F	efflux modulator	0.000	0.000	[0.000, 0.000]
R1	log10 F	formulation architecture	0.279	0.289	[0.279, 0.298]
R1	success	peptide and epithelium	0.164	0.656	[0.630, 0.690]
R1	success	enhancer pharmacology	0.137	0.656	[0.628, 0.686]
R1	success	efflux modulator	0.000	0.000	[0.000, 0.000]
R1	success	formulation architecture	0.106	0.631	[0.603, 0.659]
R1	log10 DH	peptide and epithelium	0.226	0.226	[0.219, 0.233]
R1	log10 DH	enhancer pharmacology	0.400	0.401	[0.389, 0.412]
R1	log10 DH	efflux modulator	0.000	0.000	[0.000, 0.000]
R1	log10 DH	formulation architecture	0.374	0.375	[0.363, 0.383]

Table S3c. *Group indices (closed = group alone; total = group including interactions).*

Table S4. Structural variants

Variant	P_0	P_0/base	R1 median *F*	P1 median *F*	R1 P(success)	P1 P(success)
lumped (base)	0.0094	0.965	1.20	4.77	13.4	43.8
lumped + stagnant layer	0.0267	2.73	1.20	6.14	10.1	54.7
lumped + stagnant layer, P0 not recalibrated	0.00975	1	0.70	3.43	3.9	33.4
spatial, dispersion x3	not calibratable					
spatial, dispersion x10	not calibratable					
spatial, dispersion x30	not calibratable					
spatial, dispersion x100	0.0528	5.41	1.25	14.87	14.2	73.8
spatial, dispersion x100, P0 not recalibrated	0.00975	1	0.29	4.11	3.6	39.6
spatial, dispersion x1000	0.0118	1.21	1.20	5.66	13.4	48.6
spatial, dispersion x1000, P0 not recalibrated	0.00975	1	1.00	4.79	11.1	44.2
spatial, dispersion x1e+06	0.00931	0.955	1.18	4.68	13.0	43.2
spatial, dispersion x1e+06, P0 not recalibrated	0.00975	1	1.23	4.88	13.8	44.6

Table S4. *All structural variants, including the variants in which the baseline permeability was not recalibrated (P0 kept at the lumped value). Median F and P(success) in %; P_0 in cm h^{-1}.*

Convergence and numerics

Arm	Draws	Median *F* (%)	P(success) (%)	P(*F*≥4%) (%)
R1	250	1.098	11.60	17.20
R1	500	1.118	11.20	16.40
R1	1,000	1.109	12.80	18.40
R1	2,000	1.109	13.75	20.25
R1	5,000	1.170	14.26	21.06
R1	10,000	1.175	13.85	19.87
R1	20,000	1.174	14.11	20.23
R1	50,000	1.188	14.38	20.39
R1	100,000	1.192	14.61	20.67
P1	250	4.801	50.40	58.40
P1	500	4.514	46.80	54.80
P1	1,000	4.701	45.40	55.00
P1	2,000	4.745	44.75	55.05
P1	5,000	4.852	44.12	55.58
P1	10,000	4.849	44.22	55.70
P1	20,000	4.887	44.27	55.77
P1	50,000	4.920	44.54	56.14
P1	100,000	4.953	44.75	56.26

Table S5. *Running estimates (one stream of 10^5 draws).*

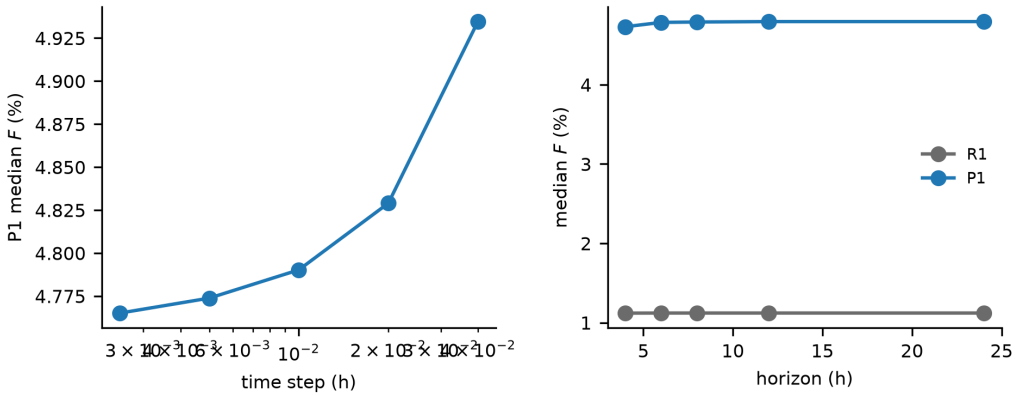


Fig. S1. *Numerical convergence of the median bioavailability: time step (left, P1) and horizon (right).*

Table S6. Verification tests

Test	What is checked
Original model (oral_peptide_mc.py --test)	peptide mass balance closes to 10^{-9}; $F \in [0,1]$; determinism; calibration of R1 within 5%; enhancer raises *F* over peptide alone; efflux module never lowers *F* and has no effect on non-substrates; time-step convergence; rank correlation of identical vectors.
Sampler equivalence	the uniform-variate sampler reproduces the original P1 success probability ($\lvert z \rvert < 3$) and median *F* (within 6%).
Mass balance (lumped)	closure error below 10^{-9} in every arm of the paired design (largest: 4×10^{-15}).
Sobol estimators	the Saltelli and Jansen estimators recover the analytic first- and total-order indices of the Ishigami function (maximum error below 0.04) and give zero for inert inputs.
Gaussian copula	realised Spearman correlation equals $(6/\pi)\arcsin(\rho/2)$ within 0.01.
Spatial model, well-mixed limit	for a very large dispersion multiplier the spatial model reproduces the lumped median *F* within 3%.
Spatial model, mass balance	closure error below 10^{-7} (observed 10^{-12}).
Spatial model, convergence	doubling the mesh and halving the time step changes the median *F* by less than 3%.
Spatial model, monotonicity	slower mixing lowers *F* for the same draws.
Stagnant layer	never raises *F* for the same draws; mass balance closes to 10^{-9}.
Module logic with the new sampler	no effect of the modulator if the peptide is never a substrate; the enhancer and modulator multiply the matrix-only median *F* by a factor of at least 4.
Determinism and helpers	identical results for a fixed seed; the paired-difference helper returns the exact contrast.

Table S6. *Automated verification tests (python oral_peptide_mc.py --test: 9 tests; python run_extended.py --test: 16 tests). They verify the software, not the biology.*

Table S7. Literature values used in the plausibility checks

Quantity	Value	Source	Ref.
Oral semaglutide (SNAC) absolute bioavailability	0.8% under the recommended dosing conditions; within-subject variability 137%; falls with larger water volume and rises with longer post-dose fasting	Overgaard et al. 2021, abstract	(Overgaard et al., 2021)
Oral semaglutide, SNAC dose	exposure highest with 300 mg SNAC among the doses tested (150 to 600 mg); exposure proportional to the semaglutide dose	Granhall et al. 2019, abstract	(Granhall et al., 2019)
SNAC mechanism	transcellular; confined to the area next to the tablet; transient; requires co-formulation; local buffering protects against degradation	Buckley et al. 2018, abstract	(Buckley et al., 2018)
Enhancer-based oral peptides, typical bioavailability	about 1%, low and variable	Brayden et al. 2020, abstract	(Brayden et al., 2020)
Engineered GLP-1 agonist with enhancers (dogs)	bioavailability about 6% (enteric-coated tablets, sodium chenodeoxycholate and propyl gallate)	Pechenov et al. 2021, abstract	(Pechenov et al., 2021)
Adherent mucus gel thickness (rat)	jejunum about 123 μm; colon about 830 μm; firmly adherent layer 20 to 154 μm (small intestine 20 μm or absent)	Atuma et al. 2001, abstract	(Atuma et al., 2001)

Table S7. *Values quoted from the abstracts of the sources (full texts were not read). The values are used only to place the model output in context and are not calibration data, except for the 1 to 2% baseline of R1.*

References

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